UNIT 1 PROJECT: Sustainable Design

PROJECT GUIDE

Project Brief

Design a **sustainable work or living space** addressing one or more of the United Nations Sustainable Development Goals (SDGs). This could be a home, office, laboratory, factory or other space in which people live or work. Your design should be clearly indicated and should incorporate a **variety of technologies**. Present your design at a **poster-display event**.

Aim

This project assesses your ability to work effectively as a team when **researching**, **designing** and **delivering** your project outputs. Successful teams will need to make use of a range of skills from technical drawing and writing to marketing and research. You will also need to be able to effectively present your designs at the showcase event.

Project Outputs

Upon completing the project your team will have produced the following:

- **Drawings** or diagrams clearly indicating your design. These could indicate your overall design or specific features. Drawings should be technical and professional and should clearly illustrate your concept. You may consider using CAD software (e.g. TinkerCAD, Sketchup, Fusion 360) however this is not required as long as any drawings and diagrams are clear and professional.
- **Poster** of your design (A2 or A3) - *physical copy required*. This should include your drawings as well as a brief explanation of key features of your design. Avoid overly long text sections. It should be attractive and informative, clearly indicating your design.
- **Design Outline** (several A4 pages) - *physical copy required*. This should include your drawings as well as thorough explanations of design features and how they address the SDGs, relevant background research and an analysis of the costs involved. Research should be supported by in-text citations and a reference list at the end of the document using Harvard style. It should be detailed while effectively communicating relevant features in an easily readable format for potential users or other interested parties.

Timeline

You have 3 weeks to **research** and **design** your sustainable work or living space and produce your **project outputs**. You will **present** your design during a poster-display event in the 4th week. Below is an outline schedule.

1st Week	Team forms, brainstorms design concepts and assigns roles. Initial research into sustainability, sustainable design and technologies.
2nd Week	Begin research into and design of specific features of your design.
3rd Week	Produce drawings, poster and design outline.
4th Week	Complete project outputs and prepare to present your poster and design outline to other students and staff during a poster-display event.

Team Roles

Once your team is formed you should assign the following roles:

- **Project Manager**: Oversees the entire project timeline and ensures all team members are on track. Coordinates meetings, sets deadlines, and ensures the final output is cohesive and meets the brief. Acts as the main point of contact for the lecturer.

- **Design Lead**: Responsible for the visual and spatial design of the work/living space. Creates drawings and diagrams in collaboration with the team. Ensures the design reflects sustainability principles and user needs.

- **Sustainability & Research Lead**: Investigates sustainable materials, technologies, and practices. Provides evidence-based recommendations to inform the design and construction approach. Overall Responsibility for production of Design Outline in collaboration with team.

- **Visual Communications Lead**: Ensures visual clarity across all outputs. Works with the team to communicate the project's key messages effectively. Overall responsibility for design and production of poster in collaboration with team.

- **Cost & Resource Coordinator**: Estimates the costs of materials, technologies and construction. Researches suppliers and sustainable sourcing options. Ensures the design is realistic and within a hypothetical budget.

Depending upon the size of your team, you may take on multiple roles or you may have more than 1 person completing each role. Remember that **ALL** team members have responsibility for production of the project outputs. If team members are not communicating / contributing, the team should email their lab leader so that we can contact them: edward.causton@ntu.ac.uk, rob.lockett2017@my.ntu.ac.uk, fiona.whittaker@ntu.ac.uk

Design Specification

Your design could be entirely new or could involve redesigning an existing space to make it more sustainable (e.g. one of the labs or other parts of the Clifton Campus). In the second case, ensure you research the existing space thoroughly and that you include sufficient re-design to warrant the project.

While no specific budget has been set, it is recommended that your team set their own hypothetical budget so that you can ensure design features are consistent (e.g. don't focus on saving £5 in one area if you spend £1 million in another!)

Your design should include the following:

- A variety of **sustainable technologies** integrated into the design.
- Consideration of **materials choices** to address the SDGs.
- Evidence of thorough **research** into relevant sustainable technologies, materials and practices.
- Evidence of **creativity** and **originality** in your design concept.
- Evidence that you have considered the **practicality** of your design for people using it.
- An attempt to research and analyze the **costs** involved in your design. *Note:* The emphasis here is on evidencing careful analysis and relevant research, not necessarily getting the 'correct' answer.

Where possible, you should make use of knowledge gained from your other Foundation Year modules. For example, if you are studying Foundation Science, you could refer to design features to minimize heat transfer in / out of the building and reduce energy use. If you are studying Introduction to Programming, you could use Python (or another programming language) to produce relevant graphs or models for use in your design outline. If you are studying Introductory Mathematics, you could perform calculations related to energy use, heat transfers or similar.

Subject Specific Considerations

As your expertise will depend upon your degree subject, you may wish to focus on the following aspects of the design:

- **Chemistry / Forensics students:** Sustainable materials use, methods to test and improve air and water quality etc.
- **Engineering / Physics students:** Energy generation methods and heat flow.
- **Computing students:** Integration of smart technologies and control systems.
- **Mathematics students:** Calculations of construction costs, surface area to volume ratios, optimizing resource use.

Assessment

- Evidence of this project should be included in your **Project Portfolio** to be submitted toward the end of the academic year (details of this will be provided in lectures). Please ensure team members have a copy of all project outputs so that they can include evidence of them in their portfolios.

- Your work will *not* be formally assessed at this stage. However, the poster-display event is a **gateway assessment** for your portfolio.

- Participation in the poster-display and contribution to your team's design is required. Failure to participate will result in a reduced grade for your portfolio.

- If there are genuine reasons why you are unable to attend, there will be opportunities at a later date to present your work individually. Information will be provided about this later in the year.
